Does Panel Regression Demonstrate Effects of Reliability, Availability, Maintainability, and Customer Experience on Telecom EBITDA

Jeffrey Hill

School of Business: Edgewood University

BUS-922-0001 Quantitative Analysis II

Dr. Kenneth Kavajecz

12/5/2025

Abstract

his study investigates whether reliability, availability, and customer-experience (RAM and CX) variables explain variation in Adjusted EBITDA among U.S. telecommunications firms. Using panel-structured data from the FCC's Measuring Broadband America program, ACSI Telecommunications Index reports, and SEC filings from 2018–2023, a fixed-effects panel regression model was estimated to assess the joint significance of twelve technical and experiential predictors. Results show that the model is statistically significant ($F = 6.56$, $p = 0.00156$), indicating that the RAM and CX predictors contribute to variation in adjusted EBITDA. Upload speed coverage relative to advertised values exhibited strong positive significance ($p < 0.01$), while packet loss and jitter showed moderate significance ($p < 0.05$). Most customer-experience indices were not significant. These findings suggest traditional reliability measures may require improved measurement or expanded datasets. Implications for telecom management and avenues for future research are discussed.

Does Panel Regression Demonstrate Effects of Reliability, Availability, Maintainability, and Customer Experience on Telecom EBITDA

Reliability, Availability, and Maintainability (RAM) emerged as formal engineering disciplines during World War II and were significantly advanced during the U.S. aerospace and space-race era. Foundational documents, *MIL-HDBK-217*, *MIL-STD-781*, and *MIL-STD-470/471*, established statistical and methodological frameworks that were later expanded by Bellcore now Telcordia (U.S. Department of Defense, n.d; Ericsson, n.d). These bodies of work form the basis of contemporary RAM methodologies adopted by IEEE and ITU-T for telecommunications infrastructure.

Customer experience (CX) is defined as the internal and subjective response of customers to any interaction with a firm (Meyer & Schwager, 2007). Within the telecommunications domain, CX encompasses purchase interactions, service performance, billing accuracy, and technical support effectiveness. These dimensions are increasingly important in a competitive environment where service differentiation is limited, and customer perceptions are shaped both by direct interactions and by indirect channels such as word-of-mouth.

Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA), popularized by John Malone in the cable industry, provides a normalized measure of operational profitability (Robichaux, 2002). It is widely used for cross-firm comparison in capital-intensive sectors such as telecommunications, where asset structure and depreciation schedules differ significantly across firms.

Question

This study investigates whether RAM and CX jointly or individually influence EBITDA among U.S. telecommunications providers. The purpose is to identify which operational and experiential variables generate competitive advantage or signal areas requiring capital investment. Previous analysis by Hill (2025) found unexpected coefficient signs, low statistical significance, and omitted formal diagnostic testing. To address these limitations, the present study re-examines the relationship between RAM, CX, and EBITDA using non-OLS estimation approaches to improve inference reliability and model robustness to answer Does Panel Regression Demonstrate the effects of RAM and CX on telecom EBITDA?

Data

Source

The data used in this study were drawn from previously compiled regulatory, industry, and financial sources. These datasets were originally aggregated and analyzed in Hill (2025) and stored in the file *2025_Module7_Deliverable_Aggregate_Telecommunications_Data_from_2018–2022.xlsx*. This is non-simulated data.

Data Structure

This dataset is a balanced panel which captures repeated annual observations for telecommunications firms included in the FCC's Measuring Broadband America program and the ACSI Telecommunications Index. Firms included are CenturyLink / Lumen Technologies, Frontier Communications, Charter Communications (Spectrum), Comcast, Altice USA (Optimum), and Verizon. AT&T had incomplete data for analysis. Time was observed over a 5-year period from 2018 to 2022. Each firm year pairing includes FCC technical quality variables (jitter, latency, packet loss, load times, download/upload speeds, ratio to advertised speeds); ACSI service indices (TV, Internet, Landline, VOD); and SEC-derived Adjusted EBITDA values.

Variable Definitions

Reliability Values:

Jitter is the variation in a timing signal from its ideal position over time, for networks this is normally in milliseconds (ITU TSS, 1996). Latency is the round-trip time it takes a packet to reach its destination and return to its source, measured in RTT milliseconds (ITU TDB, 2020). Packet loss is simply the loss of an internet protocol packet as a percentage of loss. Webpage load time is measured in milliseconds. Download and upload speed coverage in peak Megabits per second are the maximum speeds available during peak hours.

Availability Values:

Download and upload speed ratios are a percentage of the values provided to values advertised. ACSI TV Index, Internet Index, Landline Index, VOD Index are consumer comparison indices provided by American Customer Services Index (ACSI) annual reports.

Maintenance Values:

No maintenance values were used in this study. Common maintenance values are Mean time to repair (MTTR) and mean time between failure (MTBF) (ITU, 2001 & 2008; ETSI 1994). Support times and call times may also be relevant, but none of these values are normally available outside of the company.

Data Cleaning and Preparation

Data was loaded using appendix: data_load.R. Variables were named from the previous human readable forms to R readable in appendix: *clean_data.R*. Years were converted from string values to text. Company names were converted to categorical variable using R *as.factor()* function. EBITDA was converted from large value in millions to a log version to reduce skew, stabilize variance, and make residuals more homoscedastic. The transformation of ln_ebitda allows coefficients to be interpreted as percentage changes.

Methodology

Technique Panel Fixed and Random Effects Estimation

Panel regression was selected because the telecommunications performance and financial data obtained from the FCC, ACSI, and SEC consist of repeated firm-level observations across time, forming a longitudinal structure suitable for econometric panel methods (Wooldridge, 2010; Baltagi, 2013). This structure enables the identification of cross-sectional heterogeneity and temporal changes that cannot be modeled using simple cross-section OLS. The present analysis advances prior work by moving from pooled OLS with multicollinearity checks to fixed-effects (FE) panel estimation, which controls for all unobserved, time-invariant firm characteristics (Wooldridge, 2010).

The Fixed Effect model: $\ln(EBITDA_{it}) = \beta X_{it}+\alpha_i+\delta_t+\varepsilon_{it}$, where: X_{it} = RAM and CX predictors, α_i = firm fixed effect, δ_t = year fixed effect, ε_{it}= the residual error term (Wooldridge, 2010). This formulation follows established treatments of fixed-effects estimation in panel econometrics (Wooldridge, 2010). The full FE specification was implemented in R 4.5.2 using the *plm* package, with supporting scripts included in the appendix (*panel_data.R*, *simple_FE.R*, *full_FE.R*) (Croissant & Millo, 2008; OpenAI, 2025; Posit, 2025; R Core Team, 2025). Random-effects (RE) estimation and the Hausman specification test were attempted in *full_FE.R*, but the test statistics were undefined for the current dataset due to the non-invertible covariance difference matrix (Hausman, 1978). This

outcome is often observed in small-sample or highly multicollinear panels. R output demonstrating this is provided in the appendix. Additional tests for heteroskedasticity, autocorrelation, and model stability employed lmtest (Zeileis & Hothorn, 2002), tseries (Trapletti & Hornik, 2023), and sandwich robust covariance estimators (Zeileis, 2006).

Estimation Considerations

Panel regression using the *plm* package in R does not require strict covariance stationarity for the regressors. The fixed effects estimator relies on the assumption of strict exogeneity and not on stationarity, and therefore nonstationary regressors do not cause estimation failure in *plm*. For this reason, stationarity tests were not required prior to estimation. The software does not issue an error when nonstationary variables are present because the within transformation removes all time-invariant components, allowing consistent estimation of the slope coefficients even when the regressors evolve over time.

Heteroskedasticity and serial correlation are well-documented characteristics of firm-level panel data, and telecommunications performance data typically exhibit both phenomena. The fixed effects estimator remains unbiased in the presence of heteroskedasticity and serial correlation, but the associated standard errors are not valid unless they are corrected. To ensure correct inference, all hypothesis testing was performed using a cluster-robust covariance estimator of the HC1 type, clustered at the firm level. The following specification was used in the analysis:

```
vcovHC(fe_full, type = "HC1", cluster = "group")
coeftest(fe_full, vcov = vcovHC(fe_full, type = "HC1", cluster = "group"))
```

This estimator corrects the standard errors for heteroskedasticity across firms, serial correlation within firms, and the unbalanced structure of the panel. By applying these corrections, the fixed effects coefficient estimates remain consistent, and the inference becomes valid under general forms of within-group dependence (Zeileis & Hothorn, 2002; Zeileis, 2006).

Serial correlation does not bias the fixed effects estimator itself because autocorrelation affects the second moments of the error term rather than the consistency of the slope parameters. However, failure to adjust for serial correlation leads to underestimated standard errors. The clustered HC1 estimator resolves this problem

without requiring additional diagnostic testing. For this reason, separate tests such as the Wooldridge test for autocorrelation were not required for the main results, although they may be included in future robustness checks.

Random effects estimation and the Hausman test were attempted using the same dataset. The Hausman test statistics could not be computed because the difference in covariance matrices was not invertible, which is common in small-sample panels or in models with substantial multicollinearity among regressors. In such cases, econometric guidance recommends defaulting to fixed effects, which is consistent with the theoretical expectation that telecom firms differ in their structural characteristics.

Regression tables were generated using the Stargazer package to ensure that output was reproducible and professionally formatted.

Results

Regression Results

The fixed-effects panel regression estimating the influence of RAM and CX variables on ln(EBITDA) identified three statistically significant predictors. Jitter and packet loss were significant at the 5% level, and upload-speed coverage was significant at the 1% level. These metrics are mechanically related in network operations because both can be derived from standard ping diagnostics. This underlying engineering relationship is outside the scope of this study; future multicollinearity tests or variance-decomposition diagnostics may clarify whether these metrics provide distinct economic information or simply capture overlapping reliability phenomena.

Packet loss is well understood to degrade all Internet applications. Even small loss rates can disproportionately affect application-layer throughput because of TCP retransmission behavior. Jitter, by contrast, is more impactful for time-sensitive applications like SIP and gaming. Their statistical significance indicates that reliability impairments detectable at the transport or network layer may translate into measurable effects on firm-level financial performance when evaluated across providers and years.

Upload-speed coverage was the strongest predictor in the model. Although consumers often emphasize download speed in purchasing decisions, asymmetric broadband architecture constrains the upstream direction more severely. Limited upload capacity can inhibit video conferencing, home-office traffic, and modern interactive

applications. Telco consumers frequently select plans based on the perceived status or advertised capability of the service rather than on an objective match to their usage needs. As a result, violations of expected upload performance, particularly when upload speeds fail to be met during peak-hours, may produce disproportionate dissatisfaction relative to download performance shortcomings. This provides a reasonable mechanism for why upload-speed availability exhibits higher economic relevance in the model.

Collectively, these findings suggest that both reliability impairments and asymmetric-speed constraints can influence firm-level EBITDA. While RAM and CX variables do not uniformly exhibit significance, the subset that does aligns with engineering expectations and consumer-behavior theory.

Hypothesis Testing

The question, "Does Panel Regression Demonstrate the effects of RAM and CX on telecom EBITDA?", requires formal hypothesis testing. Because the fixed-effects estimator removes time-invariant firm heterogeneity, the tests focus on whether the RAM and CX covariates jointly and individually contribute to explaining variation in Adjusted EBITDA. The hypotheses are: H_0: RAM and CX variables have no effect on Adjusted EBITDA when firm-level fixed effects are controlled ($\beta_1 \ldots \beta_k = 0$). H_{1a}: At least one RAM or CX variable significantly predicts Adjusted EBITDA under a fixed-effects model ($\exists\ \beta_i \neq 0$). H_{1b}: Both RAM and CX contain variables which significantly predict Adjusted EBITDA.

The model was statistically significant overall ($F = 6.56$, $p = 0.00156$), indicating that the RAM and CX predictors jointly explain variation in EBITDA. This result alone is sufficient to reject H_0 in favor of $H_{1a.}$ Multiple variables exhibited statistically significant t-statistics. However, significance did not occur across all RAM variables nor any CX variables. CX indices exhibited predominantly non-significant coefficients. Based on the joint F-test and individual t-tests: H_0 is rejected, $H_{1a.}$ is accepted, and H_{1b} is not fully supported.

Implications

Managerial Implications

The results identify upload speed coverage during peak-hours as the most influential predictor of Adjusted EBITDA. This finding indicates that telecommunications management should prioritize initiatives that ensure upload speeds have consistently deliverable performance in the field. Upload performance is often shaped

by provisioning rules, equipment profiles, congestion along upstream paths, and oversubscription practices. Aligning marketed and delivered upload speeds reduces customer dissatisfaction, minimizes complaint volume, and protects EBITDA.

Packet loss and jitter were also statistically significant. These reliability indicators suggest that management should expand investment in real-time performance monitoring and automated remediation tools. Packet loss affects every application layer protocol and often results in service slowdowns or timeouts that customers perceive as instability. Jitter affects interactive and latency sensitive applications, including SIP voice, gaming, or live streaming. Reducing impairments preempts customer service calls and reduces complaint rates reported to the FCC.

Theoretical Implications

The results of this study demonstrate that not all RAM dimensions exert comparable economic influence. Availability showed less associations. The significance of upload speed coverage suggests that peak upload performance constitutes a structural bottleneck in modern broadband networks, making it a critical determinant of perceived service quality and customer retention. Peak-hour performance reflects a network's stress response under congestion and its ability to maintain service levels during periods of highest demand. Fixed-effects model control for structural differences across firms, the significance of upload performance indicates that variations within firms over time, rather than between firms, predict financial outcomes. Theoretically, this implies that ongoing network improvements that enhance peak upload performance translate into measurable increases in EBITDA.

Limitations

Several limitations should be acknowledged when interpreting the results of this research. First, the study relies on FCC Measuring Broadband America (MBA) data, which is collected from volunteer households using SamKnows hardware. Because participation is voluntary and skewed toward certain provider footprints, the dataset may not be fully representative of all network conditions or customer experiences. Second, financial data from SEC filings were matched to technical performance data at the firm level, but several companies in the FCC dataset are privately held and do not report EBITDA. This required the exclusion of some providers and reduced

the overall sample size. Third, the study focused on upload and download performance metrics, latency, jitter, and packet loss, but did not incorporate direct measures of outage frequency, mean time to repair (MTTR), network resilience, or maintenance activity. These variables are central to RAM theory but are not publicly available. Fourth, customer experience variables are limited to ACSI index scores. These indices exhibit limited variation across providers and years, which reduces statistical power and may contribute to their lack of significance in the results. Fifth, although fixed-effects panel regression controls for unobserved firm-specific characteristics, the model assumes time-invariant heterogeneity. Structural shifts, such as major network upgrades, mergers, or changes in consumer usage patterns, may not be fully captured. Finally, the cross-sectional nature of the FCC performance snapshots and annual financial reporting restrict insight into causal relationships. Improvements in network performance may increase EBITDA, but higher EBITDA may also enable greater capital investment that improves network performance.

Future Research Suggestions

Several avenues for future inquiry emerge from the findings and limitations of this study. First, intercorrelation among the RAM and CX variables should be formally evaluated using the correlation matrices, variance inflation factors, or principal component analysis. Reducing multicollinearity through variable clustering or dimension-reduction techniques may produce an improved model. Some are presented in the appendix.

Second, future studies should be conducted in partnership with telecommunications firms willing to provide access to customer-level operational data. Actual customer trouble reports, provisioning records, service tickets, and resolution logs would allow more precise linkage between technical faults and financial outcomes. Metrics, such as Mean Time to Repair (MTTR) and Mean Time Between Failures (MTBF), which are expected to carry substantial explanatory power for EBITDA yet are rarely disclosed publicly.

Finally, the financial dependent variable could be broadened. EBITDA provides an aggregate measure of operational performance, but additional financial metrics such as ARPU, churn rate, free cash flow, or segment-level profitability may reveal different relationships with RAM and CX factors. Conducting sensitivity analyses across alternative financial indicators would allow triangulation of which technical and experiential variables matter most in varying financial contexts.

References

American Customer Satisfaction Index LLC (ACSI LLC). (2018, May 22). Press release — Telecommunications 2018 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2018/05/22/press-release-telecommunications-2018/

American Customer Satisfaction Index LLC (ACSI LLC). (2020, June 9). ACSI telecommunications report 2019–2020 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/reports/2020/06/09/acsi-telecommunications-report-2019-2020/

American Customer Satisfaction Index LLC (ACSI LLC). (2021, June 8). Press release — Telecommunications study 2020–2021 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2021/06/08/press-release-telecommunications-2020-2021/

American Customer Satisfaction Index LLC (ACSI LLC). (2022, June 7). ACSI telecommunications study 2021–2022 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/reports/2022/06/07/acsi-telecommunications-study-2021-2022/

American Customer Satisfaction Index LLC (ACSI LLC). (2023, June 6). Press release — Telecommunications Study 2022–2023 [Press release]. Retrieved August 20, 2025, from https://theacsi.org/news-and-resources/press-releases/2023/06/06/press-release-telecommunications-study-2022-2023/

Altice USA, Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1702780/000162828019002370/a2018123110-kalticeusa.htm

Altice USA, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828020001613/atus-20191231.htm

Altice USA, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828021001975/atus-20201231.htm

Altice USA, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828022002873/atus-20211231.htm

Altice USA, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828023004614/atus-20221231.htm

Altice USA, Inc. (2025). Form 10-K: Annual report for the fiscal year ended December 31, 2024. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001702780/000162828025005486/atus-20241231.htm

AT&T Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/732717/000073271723000011/t-20221231.htm

Baltagi, B. H. (2013). *Econometric analysis of panel data* (5th ed.). Wiley.

Breusch, T. S., & Pagan, A. R. (1979). *A simple test for heteroscedasticity and random coefficient variation*. *Econometrica, 47*(5), 1287–1294. https://doi.org/10.2307/1911963

CenturyLink, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892620000009/ctl2019123110k.htm

CenturyLink, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892620000009/ctl2019123110k.htm

Charter Communications, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166723000024/chtr-20221231.htm

Charter Communications, Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1091667/000109166719000029/chtr12312018-10k.htm

Charter Communications, Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166720000024/chtr12312019-10k.htm

Charter Communications, Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166721000022/chtr-20201231.htm

Charter Communications, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001091667/000109166722000024/chtr-20211231.htm

Comcast Corporation. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1166691/000116669119000005/cmcsa-12312018x10k.htm

Comcast Corporation. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669120000008/cmcsa-12312019x10k.htm

Comcast Corporation. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669121000008/cmcsa-20201231.htm

Comcast Corporation. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669122000009/cmcsa-20211231.htm

Comcast Corporation. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001166691/000116669123000010/cmcsa-20221231.htm

Ericsson. (n.d.). *Telcordia heritage and network reliability standards*. https://www.ericsson.com/

Federal Communications Commission, Office of Engineering and Technology. (2019). Statistical averages (September 2018) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2019/Statistical-Averages-sept2018.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2020). Statistical averages (September 2019) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2020/Statistical-Averages-sept2019.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2021). Statistical averages (September 2020) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from http://data.fcc.gov/download/measuring-broadband-america/2021/Statistical-Averages-sept2020.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2022). Data processing flow (September 2021) [PDF]. Measuring Broadband America. Retrieved August 19, 2025, from https://data.fcc.gov/download/measuring-broadband-america/2022/data-processing-flow-sept2021.pdf

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (2023). Statistical averages (September 2022) [Data set]. Measuring Broadband America. Retrieved August 19, 2025, from https://data.fcc.gov/download/measuring-broadband-america/2023/Statistical-Averages-sept2022.xlsx

Federal Communications Commission, Office of Engineering and Technology (FCC OET). (n.d.). Measuring Broadband America: Open methodology. Retrieved August 19, 2025, from https://www.fcc.gov/oet/mba/measuring-broadband-america-open-methodology

Frontier Communications Parent, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000156276223000055/fybr-20221231x10k.htm

Frontier Communications Corporation. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/20520/000002052019000005/ftr-20181231x10k.htm

Frontier Communications Corporation. (2020). Form 10-K/A: Amendment to annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000114036120009993/form10ka.htm

Frontier Communications Corporation. (2021). Form 10-K/A: Amendment to annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000114036121007227/brhc10021007_10ka.htm

Frontier Communications Parent, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0000020520/000002052022000004/fybr-20211231x10k.htm

Greene, W. H. (2018). *Econometric analysis* (8th ed.). Pearson.

Hausman, J. A. (1978). *Specification tests in econometrics*. Econometrica, 46(6), 1251–1271. https://doi.org/10.2307/1913827

Hill, J.D. (2025). *Regression Analysis of US Telecom Reliability, Availability, and Maintenance (RAM) and Customer Experience (CX) Variables and Their Effects on EBITDA*. Unpublished manuscript, School of Business. Edgewood University.

Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables. R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

International Telecommunication Union (ITU). (2001). Communications quality of service: A framework and definitions (Recommendation ITU-T G.1000). https://www.itu.int/rec/T-REC-G.1000-200111-I/en

International Telecommunication Union (ITU). (2008). Definitions of terms related to quality of service (Recommendation ITU-T E.800). https://www.itu.int/rec/T-REC-E.800-200809-I/en

International Telecommunication Union (ITU). (2017). Vocabulary for performance, quality of service and quality of experience (Recommendation ITU-T P.10/G.100). https://www.itu.int/rec/T-REC-P.10-201711-I/en

International Telecommunication Union, Telecommunication Development Bureau (ITU TDB). (2020). ITU methodology for collecting administrative data for QoS indicators [PDF]. Retrieved August 20, 2025, from https://www.itu.int/en/ITU-D/Statistics/Documents/publications/handbook/QoS_2020_indicators_methodology_web.pdf

International Telecommunication Union, Telecommunication Standardization Sector (ITU TSS). (1996, August 27). Definitions and terminology for synchronization networks (ITU-T Rec. G.810 (08/96)). ITU. Retrieved August 20, 2025, from https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.810-199608-I!!PDF-E&type=items

Lumen Technologies, Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission.

https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892622000007/lumn-20211231.htm

Lumen Technologies, Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892622000007/lumn-20211231.htm

Lumen Technologies, Inc. (2024). Form 10-K: Annual report for the fiscal year ended December 31, 2023. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000018926/000001892624000016/lumn-20231231.htm

Meyer, C., & Schwager, A. (2007). *Understanding customer experience*. Harvard Business Review, 85(2), 116–126. Understanding customer experience - PubMed

OpenAI. (2025). ChatGPT (Version GPT-5.1) [Large language model]. https://chat.openai.com/

Posit Software, PBC. (2025). *RStudio* (Version 2025.09.2, Build 418) [Computer software]. https://posit.co/

R Core Team. (2025). *R: A language and environment for statistical computing* (Version 4.5.2) [Computer software]. R Foundation for Statistical Computing. https://www.r-project.org/

Robichaux, M. (2002). *Cable Cowboy: John Malone and the Rise of the Modern Cable Business.* Wiley.

Trapletti, A., & Hornik, K. (2023). tseries: Time series analysis and computational finance (Version 0.10-57) [R package]. Comprehensive R Archive Network (CRAN). https://CRAN.R-project.org/package=tseries

U.S. Department of Defense. (n.d). *MIL-HDBK-217: Reliability prediction of electronic equipment.* U.S. Department of Defense.

U.S. Department of Defense. (1966, March 21). *MIL-STD-470. Maintainability Program for Systems and Equipment.* U.S. Department of Defense.

U.S. Department of Defense. (n.d). *MIL-STD-471. Maintainability Demonstration.* U.S. Department of Defense.

U.S. Department of Defense. (n.d). *MIL-STD-781. Reliability Testing for Engineering Development, Qualification and Production.* U.S. Department of Defense.

Verizon Communications Inc. (2019). Form 10-K: Annual report for the fiscal year ended December 31, 2018. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/732712/000073271219000012/a2018q410-k.htm

Verizon Communications Inc. (2020). Form 10-K: Annual report for the fiscal year ended December 31, 2019. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271220000014/a2019q410-k.htm

Verizon Communications Inc. (2021). Form 10-K: Annual report for the fiscal year ended December 31, 2020. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271221000012/vz-20201231.htm

Verizon Communications Inc. (2022). Form 10-K: Annual report for the fiscal year ended December 31, 2021. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271222000008/vz-20211231.htm

Verizon Communications Inc. (2023). Form 10-K: Annual report for the fiscal year ended December 31, 2022. U.S. Securities and Exchange Commission. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000732712/000073271223000012/vz-20221231.htm

White, H. (1980). *A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity*. Econometrica, 48(4), 817–838.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press

Zeileis, A. (2006). Object-oriented computation of sandwich estimators. Journal of Statistical Software, 16(9), 1–16. https://doi.org/10.18637/jss.v016.i09

Zeileis, A., & Hothorn, T. (2002). Diagnostic checking in regression relationships. R News, 2(3), 7–10. https://CRAN.R-project.org/doc/Rnews/

Appendix: R Code

data_load.R

```
# Install if needed

#install.packages(c("readxl", "dplyr", "plm"))

library(readxl)

library(dplyr)

library(plm)

# Option 1: interactive file chooser

# data_path <- file.choose()

# Option 2: hard-code the path (edit to your actual path)

data_path <-

"C:/Data/BUS922/2025_Module7_Deliverable_Aggregate_Telecommunications_Data_from_2018-

2022.xlsx"

# Read the CLEANED sheet

tele_raw <- read_excel(data_path, sheet = "CLEANED")

# Quick structural check

str(tele_raw)

nrow(tele_raw)
```

clean_data.R

```
library(dplyr)

tele <- tele_raw %>%

  rename(

    year                    = Year,

    firm_id                 = `Provider Company`,
```

```
  jitter_ms                = `Reliability (Jitter)`,
  latency_rtt_ms           = `Reliability (Latency RTT)`,
  packet_loss_pct          = `Reliability (Packet Loss)`,
  webpage_load_ms          = `CX (Webpage Load ms)`,
  dl_speed_cov_peak_mbps   = `Availability (Download Speed Coverage Peak)`,
  dl_speed_ratio           = `Availability (Download Speed Sustained Peak to Advertised)`,
  ul_speed_cov_mbps        = `Availability (Upload Speed Coverage)`,
  ul_speed_ratio           = `Availability (Upload Speed Sustained Peak to Advertised)`,
  acsi_tv                  = `Maintenance (ACSI TV Service Index, yearly)`,
  acsi_internet            = `Maintenance (ACSI Internet Service Index, yearly)`,
  acsi_landline            = `Maintenance (ACSI Landline Service Index, yearly)`,
  acsi_vod                 = `Maintenance (ACSI VOD Service Index, yearly)`,
  ebitda_millions          = `EBITDA (Millions)`
 ) %>%
 mutate(
  year     = as.integer(year),
  firm_id  = as.factor(firm_id),
  ln_ebitda = log(ebitda_millions)
 )
nrow(tele)
head(tele)
```

panel_data.R

```
library(plm)
tele_panel <- pdata.frame(tele, index = c("firm_id", "year"))
```

```
pdim(tele_panel)
```

simple_FR.R

The following code was produced to test that variables would produce a panel model.

```
fe_simple <- plm(
  ln_ebitda ~ latency_rtt_ms + dl_speed_cov_peak_mbps + acsi_internet,
  data  = tele_panel,
  model = "within"   # firm fixed effects
)
summary(fe_simple)
```

full_FE.R

```
library(dplyr)
library(plm)

# tele_panel created as before
# tele_panel <- pdata.frame(tele, index = c("firm_id", "year"))

## ------------------------------------------------------------
## Logging helper
## ------------------------------------------------------------
log_message <- function(level = c("INFO", "WARN", "ERROR"), msg) {
  level <- match.arg(level)
  timestamp <- format(Sys.time(), "%Y-%m-%d %H:%M:%S")
  cat(sprintf("[%s] %s: %s\n", timestamp, level, msg))
}

## ------------------------------------------------------------
## 1. Define full model formula (you already have this)
```

```
## ----------------------------------------------------------
panel_formula_full <- ln_ebitda ~
  # Reliability (R)
  jitter_ms + latency_rtt_ms + packet_loss_pct +
  # CX – application layer performance
  webpage_load_ms +
  dl_speed_cov_peak_mbps + dl_speed_ratio +
  ul_speed_cov_mbps + ul_speed_ratio +
  # CX – satisfaction indices
  acsi_tv + acsi_internet + acsi_landline + acsi_vod +
  # Time effects
  factor(year)

## ----------------------------------------------------------
## 2. Try Fixed Effects (FE) model
## ----------------------------------------------------------
fe_full <- NULL

log_message("INFO", "Attempting to estimate FE model...")

fe_full <- tryCatch(
  {
    model <- plm(
      panel_formula_full,
      data   = tele_panel,
      model  = "within",
      effect = "individual"
    )
    log_message("INFO", "FE model estimated successfully.")
```

```
    model
  },
  error = function(e) {
    log_message("ERROR", paste("FE model estimation failed:", e$message))
    NULL
  }
)

if (is.null(fe_full)) {
  log_message("ERROR", "FE model is NULL. Cannot proceed with FE-based inference.")
} else {
  log_message("INFO", "FE model summary (non-robust SE):")
  print(summary(fe_full))
}

## ----------------------------------------------------------
## 3. Try Random Effects (RE) model
## ----------------------------------------------------------
re_full <- NULL

log_message("INFO", "Attempting to estimate RE model...")

re_full <- tryCatch(
  {
    model <- plm(
      panel_formula_full,
      data   = tele_panel,
      model  = "random",
      effect = "individual"
```

```
    )
    log_message("INFO", "RE model estimated successfully.")
    model
  },
  error = function(e) {
    log_message("ERROR", paste("RE model estimation failed:", e$message))
    NULL
  }
)

if (!is.null(re_full)) {
  log_message("INFO", "RE model summary (non-robust SE):")
  print(summary(re_full))
} else {
  log_message("WARN", "RE model is NULL. Skipping Hausman test and RE-robust SEs.")
}

## ----------------------------------------------------------
## 4. Hausman test (only if both FE and RE succeeded)
## ----------------------------------------------------------
hausman_result <- NULL

if (!is.null(fe_full) && !is.null(re_full)) {
  log_message("INFO", "Running Hausman test (FE vs RE)...")
  hausman_result <- tryCatch(
    {
      h <- phtest(fe_full, re_full)
      log_message("INFO", "Hausman test completed successfully.")
      print(h)
```

```
    h
  },
  error = function(e) {
    log_message("ERROR", paste("Hausman test failed:", e$message))
    NULL
  }
 )
} else {
 log_message("WARN", "Hausman test skipped: FE or RE model missing.")
}

## ----------------------------------------------------------
## 5. Cluster-robust SEs (FE always attempted if FE exists;
##    RE only if RE exists)
## ----------------------------------------------------------
library(lmtest)
library(sandwich)

fe_full_robust <- NULL
re_full_robust <- NULL

if (!is.null(fe_full)) {
 log_message("INFO", "Computing cluster-robust SEs for FE model (cluster = firm)...")
 fe_full_robust <- tryCatch(
  {
    vcov_fe_cluster <- vcovHC(fe_full, type = "HC1", cluster = "group")
    ct <- coeftest(fe_full, vcov = vcov_fe_cluster)
    log_message("INFO", "FE robust coefficients computed successfully.")
    print(ct)
```

```
    ct
  },
  error = function(e) {
    log_message("ERROR", paste("FE robust SE computation failed:", e$message))
    NULL
  }
 )
} else {
 log_message("WARN", "Skipping FE robust SEs: FE model missing.")
}

if (!is.null(re_full)) {
 log_message("INFO", "Computing cluster-robust SEs for RE model (cluster = firm)...")
 re_full_robust <- tryCatch(
  {
    vcov_re_cluster <- vcovHC(re_full, type = "HC1", cluster = "group")
    ct <- coeftest(re_full, vcov = vcov_re_cluster)
    log_message("INFO", "RE robust coefficients computed successfully.")
    print(ct)
    ct
  },
  error = function(e) {
    log_message("ERROR", paste("RE robust SE computation failed:", e$message))
    NULL
  }
 )
} else {
 log_message("WARN", "Skipping RE robust SEs: RE model missing.")
}
```

```
log_message("INFO", "FE/RE estimation block completed.")
```

build_regression.R

```
install.packages("stargazer")
library(stargazer)

library(sandwich)
library(lmtest)

# Cluster-robust vcov for FE (firm-level clustering)
vcov_fe_cluster <- vcovHC(fe_full, type = "HC1", cluster = "group")

# Robust coefficient table (for console inspection)
fe_full_robust <- coeftest(fe_full, vcov = vcov_fe_cluster)
fe_full_robust

# Extract robust standard errors
fe_robust_se <- sqrt(diag(vcov_fe_cluster))

stargazer(
  fe_full,
  type = "text",                    # "latex" or "html" for paper/appendix
  se = list(fe_robust_se),
  dep.var.labels   = "ln(EBITDA Millions)",
  column.labels    = c("FE (firm, year dummies)"),
  covariate.labels = c(
    "Jitter (ms)",
    "Latency RTT (ms)",
```

```
  "Packet Loss (%)",
  "Webpage Load (ms)",
  "DL Speed Coverage Peak (Mbps)",
  "DL Speed Sustained / Advertised",
  "UL Speed Coverage (Mbps)",
  "UL Speed Sustained / Advertised",
  "ACSI TV",
  "ACSI Internet",
  "ACSI Landline",
  "ACSI VOD",
  "Year 2019",
  "Year 2020",
  "Year 2021",
  "Year 2022"
 ),
 omit.stat       = c("f", "ser"),      # optional, to keep table cleaner
 no.space        = TRUE,
 title           = "Fixed-Effects Panel Regression: RAM & CX Effects on ln(EBITDA)"
)
```

descriptive_stats.R

```
summary(tele[, c(
            "ln_ebitda",
            "jitter_ms", "latency_rtt_ms", "packet_loss_pct",
            "webpage_load_ms",
            "dl_speed_cov_peak_mbps", "dl_speed_ratio",
            "ul_speed_cov_mbps", "ul_speed_ratio",
            "acsi_tv", "acsi_internet", "acsi_landline", "acsi_vod"
           )])
```

correlations.R

```
num_vars <- tele[, c(

  "jitter_ms", "latency_rtt_ms", "packet_loss_pct",

  "webpage_load_ms",

  "dl_speed_cov_peak_mbps", "dl_speed_ratio",

  "ul_speed_cov_mbps", "ul_speed_ratio",

  "acsi_tv", "acsi_internet", "acsi_landline", "acsi_vod"

)]


cor_matrix <- cor(num_vars, use = "pairwise.complete.obs")

round(cor_matrix, 2)
```

Appendix: R Results

str(tele_raw)

```
tibble [36 × 16] (S3: tbl_df/tbl/data.frame)
 $ Year                                        : num [1:36] 2022 2022 2022 2022 2022 ...
 $ Provider Company                            : chr [1:36] "CenturyLink DSL (LUMEN)" "Frontier DSL"
"Charter Cable" "Comcast Cable" ...
 $ Reliability (Jitter)                        : num [1:36] 2.23 4.81 0.38 0.35 0.52 ...
 $ Reliability (Latency RTT)                   : num [1:36] 35.3 24.3 23.7 15.7 12.4 ...
 $ Reliability (Packet Loss)                   : num [1:36] 0.0021 0.0015 0.0007 0.0007 0.0011 ...
 $ CX (Webpage Load ms)                        : num [1:36] 2905 2855 1473 1657 1224 ...
 $ Availability (Download Speed Coverage Peak)            : num [1:36] 29.1 12.7 367.8 562 234.9 ...
 $ Availability (Download Speed Sustained Peak to Advertised): num [1:36] 0.879 0.84 1.108 1.056 1.017
...
 $ Availability (Upload Speed Coverage)                   : num [1:36] 2.98 1.04 14.58 14.86 34.19 ...
 $ Availability (Upload Speed Sustained Peak to Advertised)  : num [1:36] 0.919 0.91 1.111 1.104 0.977 ...
 $ Maintenance (ACSI TV Service Index, yearly)            : num [1:36] 63 63 63 65 60 61 71 62 63 64 ...
 $ Maintenance (ACSI Internet Service Index, yearly)        : num [1:36] 60 61 63 66 59 61 72 62 57 63 ...
 $ Maintenance (ACSI Landline Service Index, yearly)        : num [1:36] 65 59 67 69 66 59 74 65 62 69 ...
 $ Maintenance (ACSI VOD Service Index, yearly)            : num [1:36] 63 63 69 67 63 63 73 63 66 66 ...
 $ EBITDA (Millions)                           : num [1:36] 6703 6250 21616 29403 3866 ...
 $ Note                                        : chr [1:36] NA "1/2 of Company EBITDA $12.5 Billion" NA NA ...
```

nrow(tele_raw)

```
[1] 36
```

pdim(tele_panel)

Unbalanced Panel: n = 9, T = 1-5, N = 36

summary(fe_simple)

Oneway (individual) effect Within Model

Call:

plm(formula = ln_ebitda ~ latency_rtt_ms + dl_speed_cov_peak_mbps +

acsi_internet, data = tele_panel, model = "within")

Unbalanced Panel: n = 9, T = 1-5, N = 36

Residuals:

Min. 1st Qu. Median 3rd Qu. Max.

-0.567062 -0.186138 -0.010698 0.148841 0.764292

Coefficients:

	Estimate	Std. Error	t-value	Pr(>|t|)	
latency_rtt_ms	-0.06842518	0.06608203	-1.0355	0.31078	
dl_speed_cov_peak_mbps	0.00075709	0.00084671	0.8941	0.38012	
acsi_internet	0.07925634	0.03231604	2.4525	0.02184	*

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares: 5.4321

Residual Sum of Squares: 3.2737

R-Squared: 0.39734

Adj. R-Squared: 0.12112

F-statistic: 5.27453 on 3 and 24 DF, p-value: 0.0061542

summary(fe_full)

Oneway (individual) effect Within Model

Call:

plm(formula = panel_formula_full, data = tele_panel, effect = "individual",

model = "within")

Unbalanced Panel: n = 9, T = 1-5, N = 36

Residuals:

Min. 1st Qu. Median 3rd Qu. Max.

-0.359023 -0.069041 -0.013675 0.091883 0.224328

Coefficients:

	Estimate	Std. Error	t-value	Pr(>|t|)	
jitter_ms	3.2843e-01	1.2423e-01	2.6436	0.022844	*
latency_rtt_ms	-1.8243e-02	5.0211e-02	-0.3633	0.723247	
packet_loss_pct	-1.7778e+02	1.0951e+02	-1.6235	0.132774	
webpage_load_ms	1.6378e-04	2.6570e-04	0.6164	0.550164	
dl_speed_cov_peak_mbps	-9.5976e-04	9.1548e-04	-1.0484	0.316955	

```
dl_speed_ratio        -1.0433e+00  1.9645e+00 -0.5311 0.605919

ul_speed_cov_mbps      4.1154e-03  9.3511e-04  4.4010 0.001061 **

ul_speed_ratio         1.1398e-01  1.9292e+00  0.0591 0.953946

acsi_tv               3.1987e-02  5.2611e-02  0.6080 0.555533

acsi_internet         -2.2133e-02  4.8030e-02 -0.4608 0.653912

acsi_landline          1.8717e-02  4.7347e-02  0.3953 0.700169

acsi_vod              1.1120e-02  3.8100e-02  0.2919 0.775823

factor(year)2019       6.4341e-02  1.7421e-01  0.3693 0.718891

factor(year)2020       2.3897e-04  1.8001e-01  0.0013 0.998965

factor(year)2021       1.5833e-01  2.7538e-01  0.5750 0.576894

factor(year)2022       1.2686e-01  2.2443e-01  0.5652 0.583264

---

Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Total Sum of Squares:    5.4321

Residual Sum of Squares: 0.51516

R-Squared:      0.90516

Adj. R-Squared: 0.69825

F-statistic: 6.56184 on 16 and 11 DF, p-value: 0.0015569
```

re_full Error

Error in swar_Between_check(estm[[2L]], method) : model not estimable: 17 coefficient(s) (incl. intercept) to be estimated but only 9 individual(s) in data for the between model necessary for Swamy-Arora random-effect model estimation

log_message for Hausmann Test

[2025-12-03 21:40:04] WARN: Hausman test skipped: FE or RE model missing.

t test of coefficients:

```
                    Estimate  Std. Error t value  Pr(>|t|)
jitter_ms           3.2843e-01  1.5686e-01  2.0938   0.06025 .
latency_rtt_ms      -1.8243e-02  1.6851e-02 -1.0826   0.30214
packet_loss_pct     -1.7778e+02  7.9293e+01 -2.2421   0.04653 *
webpage_load_ms      1.6378e-04  1.1850e-04  1.3821   0.19437
dl_speed_cov_peak_mbps -9.5976e-04  7.4173e-04 -1.2940   0.22219
dl_speed_ratio      -1.0433e+00  1.3845e+00 -0.7536   0.46693
ul_speed_cov_mbps    4.1154e-03  4.2307e-04  9.7276 9.736e-07 ***
ul_speed_ratio       1.1398e-01  1.1321e+00  0.1007   0.92161
acsi_tv             3.1987e-02  3.7511e-02  0.8527   0.41198
acsi_internet       -2.2133e-02  2.3226e-02 -0.9529   0.36110
acsi_landline        1.8717e-02  2.1790e-02  0.8590   0.40867
acsi_vod            1.1120e-02  3.7911e-02  0.2933   0.77474
factor(year)2019     6.4341e-02  1.8696e-01  0.3441   0.73723
factor(year)2020     2.3897e-04  1.8189e-01  0.0013   0.99898
factor(year)2021     1.5833e-01  2.5338e-01  0.6249   0.54480
factor(year)2022     1.2686e-01  2.3396e-01  0.5422   0.59848
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Fixed-Effects Panel Regression: RAM & CX Effects on ln(EBITDA)

==========

Dependent variable:

---

ln(EBITDA Millions)

FE (firm, year dummies)

Jitter (ms)	0.328**
	(0.157)
Latency RTT (ms)	-0.018
	(0.017)
Packet Loss (%)	-177.779**
	(79.293)
Webpage Load (ms)	0.0002
	(0.0001)
DL Speed Coverage Peak (Mbps)	-0.001
	(0.001)
DL Speed Sustained / Advertised	-1.043
	(1.384)
UL Speed Coverage (Mbps)	0.004***
	(0.0004)
UL Speed Sustained / Advertised	0.114
	(1.132)
ACSI TV	0.032

	(0.038)
ACSI Internet	-0.022
	(0.023)
ACSI Landline	0.019
	(0.022)
ACSI VOD	0.011
	(0.038)
Year 2019	0.064
	(0.187)
Year 2020	0.0002
	(0.182)
Year 2021	0.158
	(0.253)
Year 2022	0.127
	(0.234)
Observations	36
R2	0.905
Adjusted R2	0.698

Note: *p<0.1; **p<0.05; ***p<0.01

round(cor_matrix, 2)

	jitter_ms	latency_rtt_ms	packet_loss_pct	webpage_load_ms
jitter_ms	1.00	0.71	0.76	0.77
latency_rtt_ms	0.71	1.00	0.61	0.72
packet_loss_pct	0.76	0.61	1.00	0.54
webpage_load_ms	0.77	0.72	0.54	1.00
dl_speed_cov_peak_mbps	-0.62	-0.51	-0.57	-0.60
dl_speed_ratio	-0.47	-0.26	-0.35	-0.10
ul_speed_cov_mbps	-0.33	-0.55	-0.41	-0.29
ul_speed_ratio	-0.71	-0.60	-0.62	-0.46
acsi_tv	0.04	-0.07	-0.15	0.30
acsi_internet	-0.13	-0.13	-0.24	0.19
acsi_landline	-0.27	-0.08	-0.26	0.10
acsi_vod	-0.19	-0.23	-0.25	0.22

	dl_speed_cov_peak_mbps	dl_speed_ratio	ul_speed_cov_mbps	ul_speed_ratio
jitter_ms	-0.62	-0.47	-0.33	-0.71
latency_rtt_ms	-0.51	-0.26	-0.55	-0.60
packet_loss_pct	-0.57	-0.35	-0.41	-0.62
webpage_load_ms	-0.60	-0.10	-0.29	-0.46
dl_speed_cov_peak_mbps	1.00	0.28	0.50	0.29
dl_speed_ratio	0.28	1.00	-0.02	0.67
ul_speed_cov_mbps	0.50	-0.02	1.00	0.14
ul_speed_ratio	0.29	0.67	0.14	1.00
acsi_tv	0.05	0.23	0.16	0.21

acsi_internet	0.12	0.49	0.04	0.39
acsi_landline	0.10	0.59	-0.13	0.39
acsi_vod	0.00	0.60	0.04	0.57

	acsi_tv	acsi_internet	acsi_landline	acsi_vod
jitter_ms	0.04	-0.13	-0.27	-0.19
latency_rtt_ms	-0.07	-0.13	-0.08	-0.23
packet_loss_pct	-0.15	-0.24	-0.26	-0.25
webpage_load_ms	0.30	0.19	0.10	0.22
dl_speed_cov_peak_mbps	0.05	0.12	0.10	0.00
dl_speed_ratio	0.23	0.49	0.59	0.60
ul_speed_cov_mbps	0.16	0.04	-0.13	0.04
ul_speed_ratio	0.21	0.39	0.39	0.57
acsi_tv	1.00	0.89	0.75	0.71
acsi_internet	0.89	1.00	0.88	0.75
acsi_landline	0.75	0.88	1.00	0.69
acsi_vod	0.71	0.75	0.69	1.00

Appendix: ChatGPT Prompts

1. From 2025_Module7_Deliverable_Aggregate_Telecommunications_Data_from_2018-2022.xlsx CLEANED worksheet loaded in tele_raw, clean variable names and create analysis dataset.
2. Declare a panel structure using plm with panel keys firm and year.
3. Create a simple FE model to verify everything is working.
4. Define the full panel model with year factor using Reliability variables: jitter_ms, latency_rtt_ms, packet_loss_pct, which reflect packet delay variation, round-trip delay, and packet integrity; user-perceived throughput variables of webpage_load_ms, dl_speed_cov_peak_mbps, dl_speed_ratio, ul_speed_cov_mbps, ul_speed_ratio; customer satisfaction Indices of acsi_tv, acsi_internet, acsi_landline, and acsi_vod.
5. Run Random Effects (RE) model with year factor and perform Hausman test: FE vs RE.
6. Build a Regression Table (with Robust SEs).
7. Perform basic diagnostics for Outliers / extreme values, multicollinearity, and variable scaling.